



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

PAPER 3

9188/3

NOVEMBER 2015 SESSION

50 minutes

Additional materials:

Answer paper

Electronic Calculator and / or Mathematical tables

Ruler (mm)

TIME 50 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **three** questions.

Question 1 is compulsory.

Answer any other **two** from the remaining questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Candidates are advised to spend 25 minutes on **question 1**.

This question paper consists of 7 printed pages and 1 blank page.

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[Turn over

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

refractive index,

$$n = \frac{1}{\sin C}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = 6\pi\eta r v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

Data

speed of light in free space,	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant,	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant,	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant,	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall,	$g = 9.81 \text{ m s}^{-2}$

Answer question 1 and any other 2 from the remaining questions.

- 1 (a) (i) State the base unit of temperature.
- (ii) Boltzmann's constant, k , and Planck's constant, h , are related to Stefan's constant, δ , by the expression

$$\delta = \frac{ak^4}{h^3c^2},$$

where a is a dimensionless constant and c is the speed of light.

Determine the base units of δ .

- (iii) State the limitation of using base units in the equation in (ii).

[5]

- (b) (i) Define the terms *accuracy* and *precision*.
- (ii) State and explain the error associated with precision.
- (iii) The following results in Nkg^{-1} were obtained in an experiment to determine the earth's gravitational field strength, g :

8.6; 8.7; 8.8; 8.5; 8.2; 8.9

Discuss whether the results were precise and/or accurate.

[6]

- (c) Fig.1.1 shows part of a car moving at 15 ms^{-1} on a level road. The wheel is of radius 0.30 m and has an angular velocity, ω , of 50 rad s^{-1} .

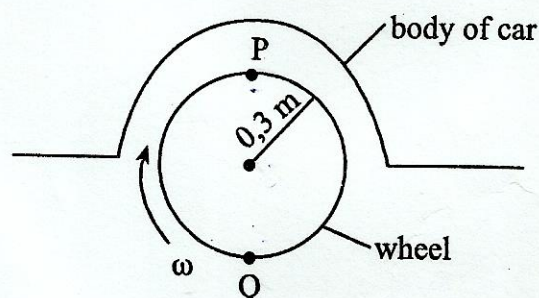


Fig.1.1

- (i) Determine the minimum co-efficient of friction to avoid skidding given that the car is going round a curve of radius 200 m at 15 ms^{-1} .

- (ii) Point P is observed as moving at twice the velocity of the centre of the wheel while point Q is observed as momentarily at rest.

Explain these observations.

[5]

- (d) The system shown in Fig.1.2 is in equilibrium.

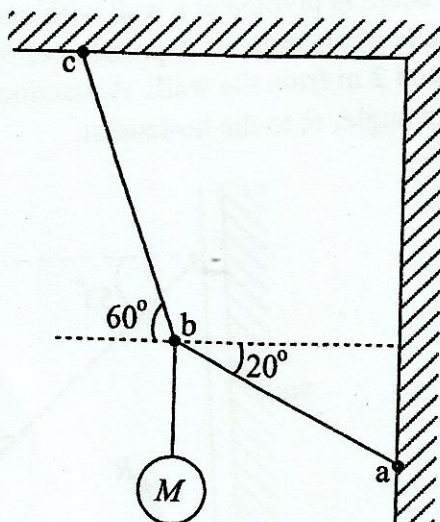


Fig.1.2

If $M = 2.0 \text{ kg}$, find the magnitude of the tension in

- (i) string ab,
- (ii) string bc.

[4]

2

- (a) Distinguish between a vector quantity and a scalar quantity.
- (b) A car travelling at 20 ms^{-1} due East turns and continues at 15 ms^{-1} due South.
 - (i) Draw a vector diagram to illustrate the
 1. initial and the final velocity,
 2. change in velocity.
 - (ii) Calculate the change in the
 1. speed of the car,
 2. velocity of the car.

[2]

[6]

- (c) Explain why the acceleration of an object may not be in the same direction as the velocity of the object.

[2]

- 3 (a) (i) Define a couple.

- (ii) State two conditions for a body to be in equilibrium.

[3]

- (b) Fig.3.1 shows a horizontal uniform beam of length 8 m and weight 200 N. The beam is pivoted at a wall with one of its ends connected to a metal cable of an angle of depression of 53° . A mass weighing 600 N is placed 2 m from the wall. A reaction force, R , acts at the pivot, P , at an angle, θ , to the horizontal.

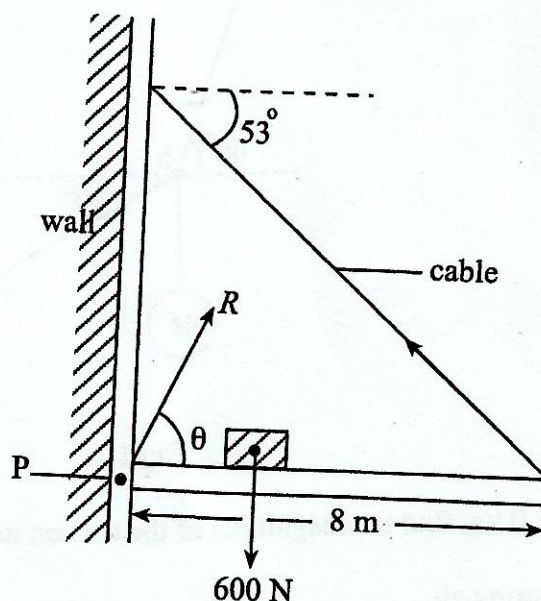


Fig.3.1

Calculate the

- (i) tension in the cable,
 (ii) reaction force, R ,
 (iii) angle θ .

[6]

- (c) Give one example of everyday application of forces in equilibrium.

[1]

- 4 (a) Define gravitational field strength.

[1]

- (b) A satellite is in a circular orbit of radius, r , about the earth of mass, M_E .

- (i) Deduce an expression for the period of the satellite.

- (ii) A satellite of mass 500 kg is in circular orbit at a height of 2 000 km above the earth's surface. The radius of the earth is 6 400 km.

Calculate the

1. gravitational field strength at the height of 2 000 km,
2. orbital speed of the satellite. [7]

- (c) Suggest why a geostationary satellite has a short lifespan. [2]